

Technical Design Document: Flue Agent Migration

Project: Agent-Native Strength & Conditioning App (Cloudflare Stack)

1. Executive Summary

This document outlines the migration from a legacy SDK approach to the **Flue Agent Framework**. The architecture leverages Cloudflare Workers, R2, D1, and Vectorize to create a multi-tenant, "Shadow Coach" system that is both reactive to user input and proactive in athlete management.

2. Data Strategy & Memory Model

To optimize context window usage and cost-efficiency, memory is partitioned into three distinct tiers:

Tier	Storage	Content Type	Access Method
Structured (Hot)	Cloudflare D1	Workout logs, sets, reps, actuals.	Kysely / TypeScript Skills
Contextual (Soul)	Cloudflare R2	User personality, "limiters", coaching tone.	Flue Virtual Sandbox
Semantic (Warm)	Vectorize	Domain knowledge (Concurrent training, etc).	RAG / Search Skills

3. Implementation Phases

Phase 1: Multi-Tenant Sandbox & Roles

Establish a secure, isolated environment for each athlete.

- Create `.flue/roles/coach.md`: Migrating "Gym Slang" and personality guidelines.

- **Security Fix:** Dynamic Sandbox Initialization. Mount the R2 bucket to a specific user prefix.

```
// .flue/agents/coach.ts
const userPrefix = `users/${userId}/`;
const sandbox = getVirtualSandbox(env.BUCKET, { root: userPrefix });
```

Phase 2: Hybrid Interaction Engine

Support both **Reactive** (User-led) and **Proactive** (Agent-led) flows.

- **Reactive:** Use Streaming Responses in Workers to prevent timeouts during complex reasoning.
- **Proactive:** Implement "Shadow Coach" logic using **Cloudflare Durable Object Alarms** or **Queues**.
- **Jitter Strategy:** Randomize execution times (e.g., `now() + random(1hr)`) to stagger LLM API calls and D1 load.

Phase 3: TypeScript Skill Adapters (The Kysely Bridge)

Since the Virtual Sandbox is shell-based, a TypeScript adapter is required to interact with the D1 database.

- Create `.agents/skills/log_set_actuals/` directory.
- Define `index.ts`: Imports Kysely services to execute `update_set_data`.
- Define `SKILL.md`: Instructs the agent on how/when to trigger the DB write.

4. Infrastructure & Security Guardrails

Auditability Note: Every D1 write initiated by an agent Skill must be logged in a `CoachInteractions` table with a snapshot of the context used for the decision.

- **Auth Handoff:** Validate Redwood-issued JWTs in the Flue Worker. The sub claim must define the R2 sandbox scope.
- **Rate Limiting:** Use Cloudflare Queues with `delay_seconds` to smooth spikes in LLM usage during peak workout hours.

- **Environment:** Store ANTHROPIC_API_KEY and other secrets in Cloudflare Worker Secrets (not in code).

5. Deployment Workflow

```
# Development
flue dev --target cloudflare --env .env

# Build & Production Deployment
flue build --target cloudflare
wrangler deploy
```